P425/2
PRINCIPAL
MATHEMATICS
Paper 2
July/August 2026
3 hours

WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

- *This examination paper consists of* **two** *sections;* **A** *and* **B.** *It has* **five** *items.*
- Section A *consists of* **one** *compulsory item.*
- Section B *has two parts* **I** *and* **II,** *each with two optional items. Respond to* **only two** *items from section B, choosing* **one** *item from* **each** *of the parts,* **I** *and* **II.**
- *Respond to* **three** *items in* **all.**
- *Any additional item(s) responded to will* **not** *be scored.*
- *Begin each item response on a* **fresh** *page.*
- *All necessary working must be shown clearly.*
- *Where necessary, use* $g = 9.8ms^{-2}$.
- *Graph paper is provided.*
- *Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.*

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SECTION A

*The item in this section is **compulsory.***

Item 1

Mr. Mukasa, a farmer in Mubende, a district 150 km from Kampala intends to start a fish farming project. The government through the Parish Development Model (PDM), provides free fish to qualified farmers.

The P.D.M officials require that for a farmer to qualify, they must have a pond with surface area of at least 110 m^2.

The farmer's only piece of land that he wants to use for this project is irregular in shape, and he intends to find out if it meets the official's requirements.

He contacted his engineer from Kampala who is to travel by public means. The engineer left Kampala at 8:00 am. On his way, he noticed that it took him 1.5 hours to reach Mityana town, which is 80 km from Kampala. He reached Kassanda town, which is 120 km from Kampala at 10:30 am.

He is expected to reach Mubende by 12:00 pm before Mr. Mukasa leaves for other duties.

The engineer established the surface area of the land by measuring it's width at intervals of 4 m along its 24 m length. The measurements are as shown below;

Distance from one end (x m)	0	4	8	12	16	20	24
Width of the pond (y m)	3.0	4.2	5.1	5.8	5.4	4.6	3.5

When the officials arrived, they accurately measured the area of the piece of land as 115 m^2.

Mr. Mukasa would consider the engineer's estimation "excellent" if it had a relative error not greater than 0.02.

Task

(a) Establish whether the engineer found Mr. Mukasa on site.

(b) Based on the engineer's estimation, decide whether Mr. Mukasa would qualify for the P.D.M funds.

(c) Was the engineer's work rated "excellent"?

SECTION B

Respond to **only two** *items from this section choosing* **one** *item from each of the parts* **I** *and* **II.**

PART I

Item 2

A fitness coach at a fitness club conducted research to investigate the training outcomes of his clients. He found out that, on a given day, the average chance of a client using the chest press machine, the leg press machine, and the rope are 0.2, 0.3 and 0.6 respectively.

The coach intends to recommend purchasing more equipment if the chance that a client uses at least two of the afore mentioned machines exceeds 65%.

He then selected 12 of his clients to participate in an expansion study, on condition that the chance that at least 4 of these clients used the leg press machine on a given day exceeds 52%.

Furthermore, he investigated the relationship between number of hours spent exercising and individual body weight. The data collected from 8 of his clients is as shown below;

Client	A	B	C	D	E	F	G	H
Hours spent exercising	1.0	1.2	1.5	1.8	1.9	2.0	2.2	2.4
Body weight (kg)	95	90	80	82	78	64	68	59

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The club will increase the exercise hours for the clients if the research results show a correlation coefficient that is significant at the 1% level.

Task

(a) Make a recommendation regarding the purchase of more pieces of equipment for the fitness club.
(b) Decide whether there is a need to expand the training facility.
(c) Make an informed decision regarding the hours of exercise.

Item 3

The management of an internet service company wants to improve the speed at which customers support calls are answered.

The table below shows the number of customers who waited for their calls to be answered during a given day.

Waiting time (minutes)	Number of customers
0 - 4	8
5 - 9	14
10 - 14	22
15 – 19	26
20 – 24	18
25 – 29	10
30 - 34	7
35 - 45	5

The company wants to determine the most common waiting time. If this time is greater than 15 minutes, management will immediately recruit more customer service agents. Additionally, the data collected is considered realistic if the standard deviation is at most 11 minutes.

The collected data shows that these waiting times follow a normal distribution. The board also wants to determine the 95% confidence interval for the mean waiting time. If the confidence interval lies entirely between 10 and 18 minutes, the customer service department will receive a pay rise.

Task

Based on appropriate mathematical calculations and justifications, advise the management on whether to:

(a) recruit more customer service agents.
(b) treat the data as realistic.
(c) make a decision regarding the pay rise.

PART II

Item 4

An officer working with Uganda National Roads Authority (UNRA) is responsible for ensuring safety on Uganda's major high ways.

One morning, the traffic control centre received information that two fuel tankers are approaching the same section of the Kampala – Jinja highway, a busy route used to transport petroleum products from Jinja to Kampala.

At 9:00 am, truck A is at position $\begin{pmatrix}0\\0\end{pmatrix}$ km and is travelling East at a constant speed of 60 $kmhr^{-1}$.

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At the same time, truck B is at position $\begin{pmatrix}15\\8\end{pmatrix}$ km relative to the fixed point and is travelling South at a constant speed of 48 $kmhr^{-1}$.
The road safety department requires an emergency warning to be issued by the engineer whenever two vehicles are expected to come within 2 km of each other.
After arriving safely at a fuel depot, truck B must unload 4 tonnes of packaged lubricants packed in 200 kg batches.
One batch at a time is pushed up an 8m smooth ramp, inclined at 25^0 to the horizontal, using a constant force of 1000 N parallel to the incline.
The depot manager has allowed 2 minutes to complete the unloading before the next delivery truck arrives.

Task
(a) Establish whether the emergency warning should be issued by the engineer.
(b) Confirm whether the unloading process of truck B will interrupt the next delivery.

Item 5
A telecommunications Company is installing a new signal mast in a rural area of Uganda. The mast is held in equilibrium by five supporting steel cables attached at different points, pulling in different directions simultaneously.
The forces in the tables are as summarized below;

Cable	1	2	3	4	5
Force (KN)	14	9	18	11	8
Direction	Due East	$N40^0E$	Due West	Due South	$S30^0E$

Safety regulators state that if the resultant force on the mast exceeds 12 kN, the structure is considered unstable and must be reinforced immediately.
After installation, a fuel truck must ascend a steep access road leading to the site.
The steepest section of the road is inclined at 12^0 to the horizontal.
To ensure continuous motion uphill, the tanker's engine provides a constant driving force of P = 42 KN.
The coefficient of friction between the tyres and the road surface is $\mu = 0.25$.
The steep section of the road is 250 m long, and the tanker enters it with an initial speed of 6 ms^{-1}. The tanker and its load have a total mass of 1800 kg.
The vehicle's transmission system is designed to safely operate up to a maximum speed of $100ms^{-1}$ under a heavy load.
If this limit is exceeded, there is a risk of mechanical failure, which could cause the tanker to lose control on the steep section. At the same time, a worker operating a crane accidentally dislodges a metal tool from a platform at the top of the cliff, which is 50 m above the road.
The tool leaves the platform with an initial speed of 20 ms^{-1} at an angle of 35^0 above the horizontal, directed towards the road where the tanker is moving.
The horizontal distance from the base of the cliff to the tanker's entry point on the road is 45 m.

Task
As a site engineer;
(a) establish whether there is a need for structural reinforcement.
(b) determine whether the tank will ascend without a mechanical risk.
(c) analyse whether the falling tool can cause danger to the tanker.

END

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